

Bayesian Analysis of CDC Wastewater Viral Activity for SARS-CoV-2, Influenza, and RSV

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Abstract

The CDC (Centers for Disease Control) collects data on viral levels in wastewater across all U.S. states and territories. This program, known as the NWSS (National Wastewater Surveillance System) provides the infrastructure to monitor the spread of communicable diseases via wastewater analysis. Three major concerns are the concentrations of SARS-CoV-2, RSV, and Influenza-A. In this project, I will present an analysis of the relationship between sample site, and viral concentration. Additionally, I will explore the effect of population size on relative share of the measured pathogens.

1 Data

Sourcing Data

The wastewater dataset was sourced from Centers for Disease Control and Prevention (2026). The data was first downloaded on April 4, 2026 and was downloaded again to update models on April 18, 2026. There are 1,279 participating test sites across U.S. states and territories Centers for Disease Control and Prevention (2025). These sites can be categorized within counties within U.S. states and cover roughly 43% of the population. The data is collected as a partnership between State & Local Health departments, CDC contracts with Verily Life Sciences, and WastewaterSCAN which is a partnership between Stanford University, Emory University, and Verily.

An example of how the data is collected is provided by the WastewaterSCAN program WastewaterSCAN (2025). The water is collected directly from wastewater treatment plants and shipped to labs at Verily. The measurements are taken as composites of wastewater samples over multiple hours and analyzed with PCR assays to target genomic pathogens.

Preparation and Normalization

The unique genetic sequence associated with each pathogen is targeted. The methodology can vary slightly between states and geography, but a trimmed average target (PMMoVw,d for day d) is calculated as follows WastewaterSCAN (2025):

State Aggregated Trendline:

$$\frac{\text{Trimmed Target}}{PMMoV} = \frac{\sum_{i=1}^x pop_i \cdot \frac{\text{Trimmed Target}_{i,d}}{PMMoV_{i,d}}}{\sum_{i=1}^x pop_i}$$

National Aggregated Trendline:

$$T_{w,r,d}^{\text{trimmed}} = \frac{\sum_{s=1}^S \text{pop}_s \cdot \frac{\text{Trimmed Target}_{s,d}}{PMMoV_{s,d}}}{\sum_{s=1}^S \text{pop}_s}$$

National trends are calculated using the trimmed and smoothed average for each state included as well as the population from each state.

There is not uniform consistency between sources for this data. For example, Verily uses the same lab infrastructure as WastewaterSCAN Verily (2023), but at the state/territory level the methods can vary Centers for Disease Control and Prevention (2024). The CDC provides guidance for municipal testing, but discretion is retained for individual approaches.

The CDC takes the following steps for data preparation:

1. Group the Data by virus, site, laboratory, and laboratory method.
2. Validate the data with long transforms, and removed outliers.
3. Determine low levels by site by looking back two years in the past.
4. Compare current levels to baselines using a normalization factor. The CDC Normalizes the measurements using the WVAL, which compares the current amount of virus to the baseline, and divides this value by the standard deviation. The larger the WVAL value is, the further away from low levels it is Centers for Disease Control and Prevention (2024).

$$WVAL = e^{\frac{x - \mu_{\text{baseline}}}{\sigma_{\text{baseline}}}}$$

5. Identify averages and medians.
6. Categorize the Values using the below table (Table 1)

Table 1: WVAL Category Thresholds by Pathogen

Pathogen	Very Low	Low	Moderate	High	Very High
Influenza A	≤ 2.7	(2.7, 6.2]	(6.2, 11.2]	(11.2, 17.6]	> 17.6
COVID-19	≤ 2.0	(2.0, 3.4]	(3.4, 5.3]	(5.3, 7.8]	> 7.8
RSV	≤ 2.5	(2.5, 5.2]	(5.2, 8.0]	(8.0, 11.0]	> 11.0

Cleaning

Though the CDC dataset was partially prepared, before modeling the data required some routine cleaning steps. The raw distribution of WVAL values showed extreme outliers with a mean of 1.2M and a maximum of 2.9×10^{11} , which is not possible for a normalized metric as it would be more than 4 standard deviations above the baseline. The 99th percentile on the log scale was 3.91, where $e^{3.91} = 49.89$, therefore, the outliers were determined by filtering > 50 .

Values above 50 were considered implausible measurements and truncated (Table 2).

Table 2: Top 10 Highest Site_WVAL Observations (Pre-Cleaning)

State	Site	Source	Pathogen	Site_WVAL
Maine	ID:724	CDC Verily	RSV	2.91×10^{11}
Maine	ID:724	CDC Verily	RSV	1.72×10^{11}
Maine	ID:724	CDC Verily	RSV	9.14×10^{10}
Michigan	ID:846	State Territory	Influenza A	3.09×10^{10}
Maine	ID:724	CDC Verily	RSV	2.18×10^9
Virginia	ID:2003	State Territory	Influenza A	6.91×10^8
Kentucky	ID:2239	State Territory	Influenza A	4.93×10^8
Virginia	ID:2014	State Territory	Influenza A	3.13×10^8
Michigan	ID:846	State Territory	Influenza A	1.40×10^8
Kentucky	ID:2239	State Territory	Influenza A	8.27×10^7

A potential source of bias in the data were sites with fewer observations. There were a number of sites with ≤ 10 observations which may produce unreliable mean WVAL.

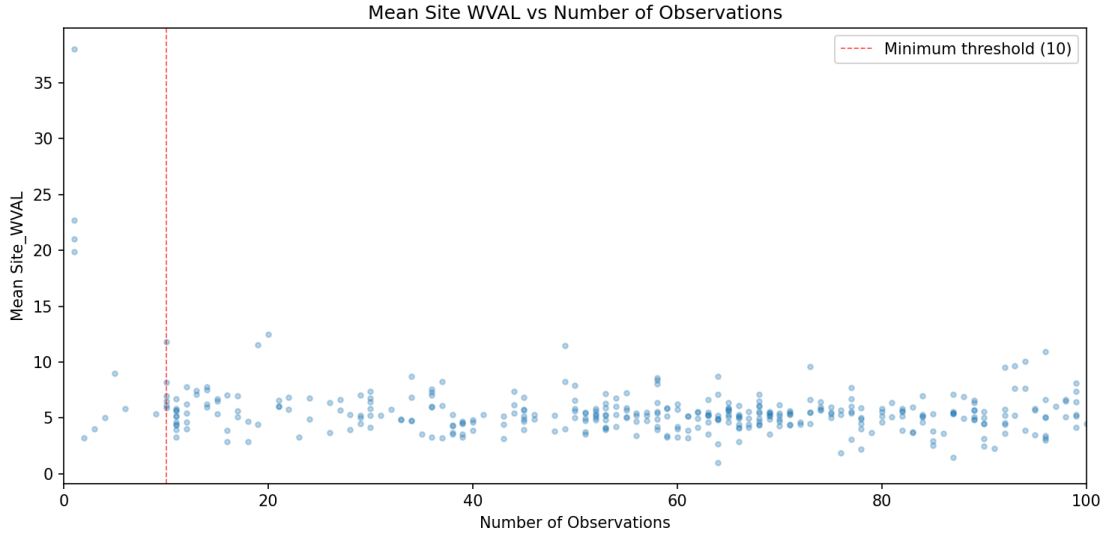


Figure 1: Mean Variability by Observation Count

In examining the pathogens measured within each site, it was discovered that some sites report zero counts for pathogens. This creates boundary values in the distribution. For model two, the data was filtered to sites that included all three pathogens, which was roughly 60.28% of sites or 1,044 of 1,732 sites.

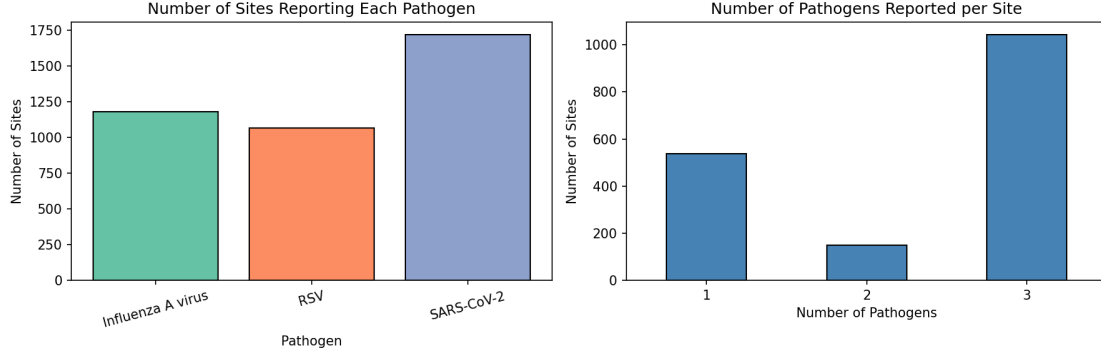


Figure 2: Pathogen Coverage

2 Methods

Model 1

The first model seeks to explore the relationship between Site ID, State/Territory, and mean WVAL. This was approached using a hierarchical (Bayesian) linear regression. The model type selection was informed by the naturally hierarchical structure within the data: National Level, State/Territory Level, and Site Level granularity are embedded into the WVAL measurements.

The model in full is written as follows

$$\begin{aligned}
y_i &\sim \mathcal{N}(\mu_i, \sigma_{site}) \\
\mu_i &= \mu_{global} + \beta_{pop} \cdot x_i + \delta_{state[i]} + \delta_{site[i]} \\
\delta_{site} &= \delta_{site,raw} \cdot \sigma_{site} \\
\delta_{site,raw} &\sim \mathcal{N}(0, 1) \\
\sigma_{site} &\sim \text{HalfNormal}(1) \\
\delta_{state} &= \delta_{state,raw} \cdot \sigma_{state} \\
\delta_{state,raw} &\sim \mathcal{N}(0, 1) \\
\sigma_{state} &\sim \text{HalfNormal}(0.5) \\
\mu_{global} &\sim \mathcal{N}(5, 1) \\
\beta_{pop} &\sim \mathcal{N}(0, 0.5)
\end{aligned}$$

where y_i is the mean Site_WVAL for site i , μ_{global} is the global intercept representing the average viral activity level across all sites, β_{pop} is the effect of log-normalized population, $\delta_{state[i]}$ and $\delta_{site[i]}$ are the state and site-level random offsets respectively, and σ_{state} , σ_{site} are the variance parameters which describes the relative contribution of each geographic level to overall variation on Site_WVAL.

The prior selection reflects what is known about the WVAL scale Centers for Disease Control and Prevention (2025). We know based on the assigned categories that the typical concentration would fall between 1-8. Given this information, a μ prior (global mean) $\mathcal{N}(5, 1)$ orients most of the mass within low to moderate

ranges.

HalfNormal(1) was chosen for state-level variance, reflecting the assumption that variation between states would be larger than variation between sites. HalfNormal(0.5) was chosen for site-level variance as a more conservative prior.

Model 2

Model 2 seeks to analyze the relationship between population, and the viral concentrations (WVAL) of three different pathogens. I hypothesized that more concentrated population center could have overall higher WVAL, but that the population size would have variable effect on individual pathogen WVAL.

A natural fit for this analysis is the Dirichlet distribution Gelman et al. (2013); PyMC Developers (2024). This distribution is a multivariate generalization of the Beta distribution up to $k - 1$ outcomes.

The PDF is

$$f(\mathbf{x} \mid \boldsymbol{\alpha}) = \frac{1}{B(\boldsymbol{\alpha})} \prod_{k=1}^K x_k^{\alpha_k - 1}$$

with normalizing constant

$$B(\boldsymbol{\alpha}) = \frac{\prod_{k=1}^K \Gamma(\alpha_k)}{\Gamma\left(\sum_{k=1}^K \alpha_k\right)}$$

The support for this distribution is a set of K-dimensional vectors that are real numbers on the interval $[0, 1]$ such that they sum to 1, i.e.

$$x_i \in [0, 1] \text{ for all } i \in \{1, \dots, K\} \text{ and } \sum_{i=1}^K x_i = 1$$

This is consistent with the data being modeled as the pathogens proportionately contribute to the total WVAL for each site.

The model in full is written as follows

$$\begin{aligned} y_i &\sim \text{Dirichlet}(\alpha_i) \\ \alpha_i &= \exp(\mu_{global} + \beta_{pop} \cdot x_i) \\ \beta_{pop} &\sim \mathcal{N}(0, 0.5) \\ \mu_{global} &\sim \mathcal{N}(0, 1) \end{aligned}$$

where the global μ is represented with a flat $\mathcal{N}(0, 1)$ without any prior information, the population β_{pop} is represented with a slightly narrowed prior $\mathcal{N}(0, 0.5)$ (modest expected effect) the concentration α is represented by an Exponential function $\alpha_i = \exp(\mu_{global} + \beta_{pop} \cdot x_i)$ (the log link ensures the concentration parameter remains strictly positive, as required by the Dirichlet).

3 Discussion

Model 1 Results

The hierarchical location model demonstrated efficient sampling with consistently high *ess_bulk* for each parameter, and convergence to $\hat{r} = 1.0$. The β_{pop} HDI excludes zero (HDI 3%: 0.028, HDI 97%: 0.153), indicating a measurable positive effect of population size on WVAL. The σ_{site} mean was 0.379 higher than the σ_{state} indicating that between-site variance had more of an impact on WVAL than between-state variance. This contradicted my initial assumption about variance. I assumed that guidance on testing being discretionary at the state level would induce more of an effect between states, as more of a standardization could exist at the state level across all of its sites.

Table 3: Posterior Summary: Hierarchical Location Model

Parameter	Mean	SD	HDI 3%	HDI 97%	MCSE Mean	MCSE SD	ESS Bulk	$\hat{r}$
μ_{global}	5.229	0.085	5.066	5.388	0.001	0.001	5242	1.0
β_{pop}	0.089	0.033	0.028	0.153	0.000	0.000	13862	1.0
σ_{state}	0.531	0.070	0.405	0.665	0.001	0.001	5419	1.0
σ_{site}	0.910	0.016	0.880	0.939	0.000	0.000	16807	1.0

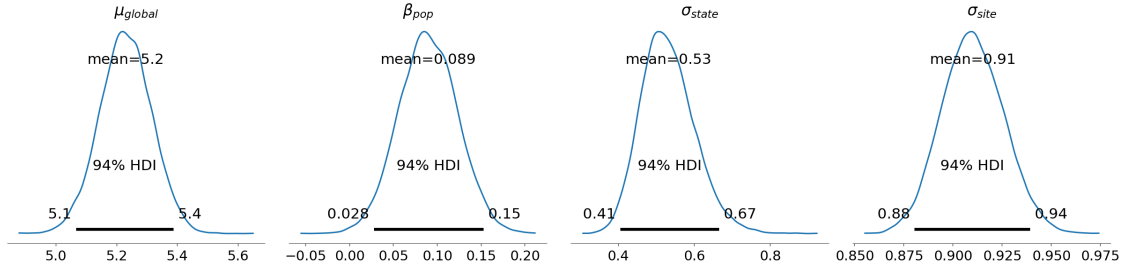


Figure 3: Posterior: Hierarchical Location Model

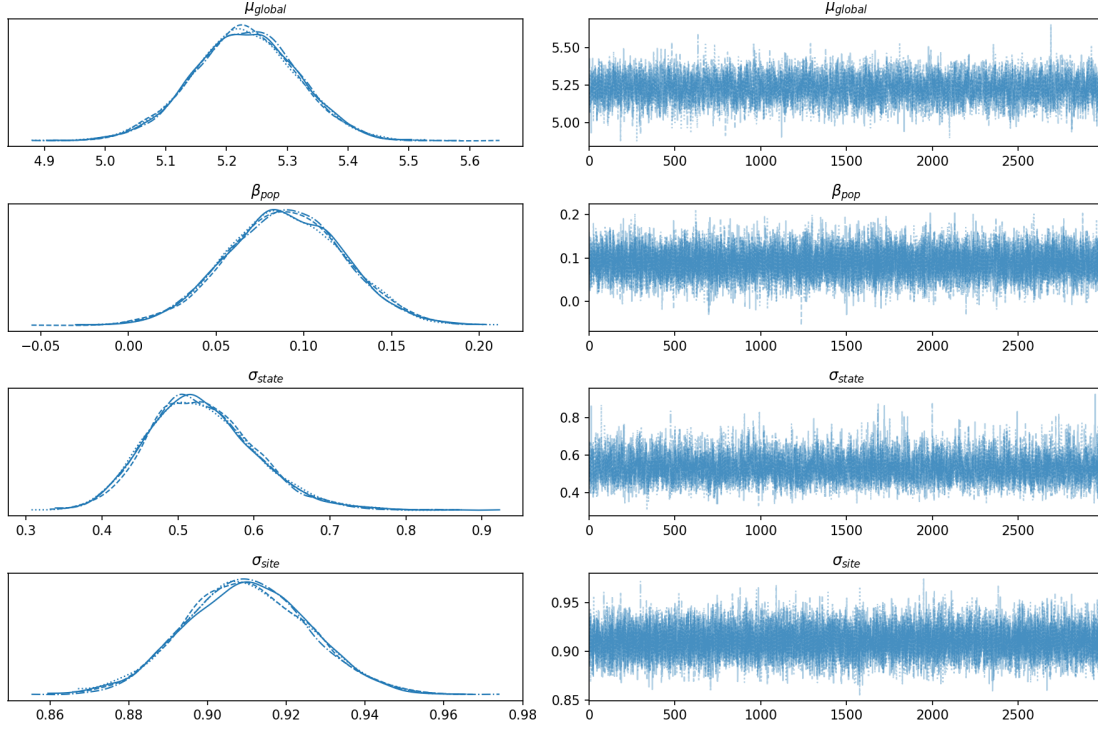


Figure 4: Trace: Hierarchical Location Model

The prior predictive summary (Table 4) shows that the μ_{global} prior covers the central plausible range of values from the WVAL scale (0-17). I postulated that a range from 1-8 is sufficient for modeling based on the categorical information provided. The σ_{state} HDI (0.000 to 1.992) allows for larger state-level variation, consistent with the prior assumption. The σ_{site} HDI covers a range that is sufficient and also allows for no variance between sites, which is plausible and acceptable in this case. In some cases, an HDI range bounded by 0 would indicate a potentially weak effect but realistically this sort of variance could happen.

Table 4: Prior Predictive Summary: Hierarchical Location Model

Parameter	Mean	SD	HDI 2.5%	HDI 97.5%
μ_{global}	4.949	0.987	3.080	6.911
σ_{state}	0.808	0.608	0.000	1.992
σ_{site}	0.397	0.294	0.004	0.950

Model 2 Results

The Dirichlet model that covers pathogen prevalence demonstrated effective sampling, with high ess_{bulk} for each parameter. The HDIs all exclude zero, indicating some effect from each parameter. The posterior mean is highest in β_{pop} [Influenza A], indicating that Influenza-A is more impacted by population size than other pathogens. The μ_{global} parameters represent baseline log-concentrations for each pathogen in the model. SARS-CoV-2 has the highest value (2.755) which corresponds to a concentration of $\exp(2.755) \approx 15.7$, reflecting a higher share of viral activity across sites compared to Influenza A (≈ 10.1) and RSV (≈ 9.3).

Table 5: Posterior Summary: Dirichlet Pathogen Prevalence Model

Parameter	Mean	SD	HDI 3%	HDI 97%	MCSE Mean	MCSE SD	ESS Bulk	$\hat{r}$
μ_{global} [Influenza A]	2.311	0.031	2.252	2.370	0.000	0.0	3926	1.0
μ_{global} [RSV]	2.234	0.031	2.176	2.293	0.001	0.0	3831	1.0
μ_{global} [SARS-CoV-2]	2.755	0.031	2.697	2.814	0.001	0.0	3795	1.0
β_{pop} [Influenza A]	0.148	0.030	0.091	0.203	0.001	0.0	3400	1.0
β_{pop} [RSV]	0.128	0.030	0.072	0.183	0.001	0.0	3445	1.0
β_{pop} [SARS-CoV-2]	0.092	0.031	0.032	0.148	0.001	0.0	3405	1.0

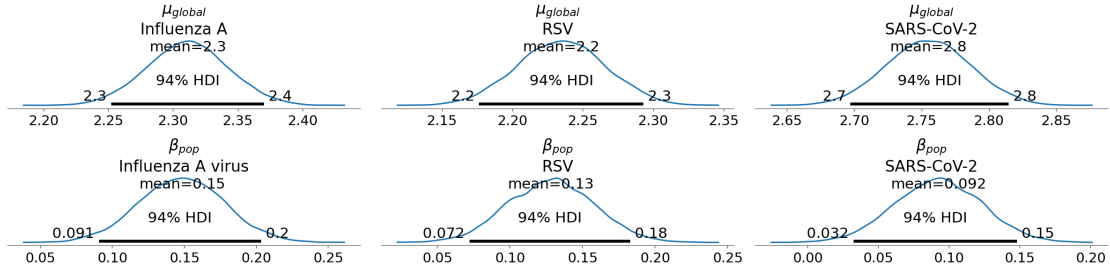


Figure 5: Posterior: Dirichlet Pathogen Prevalence Model

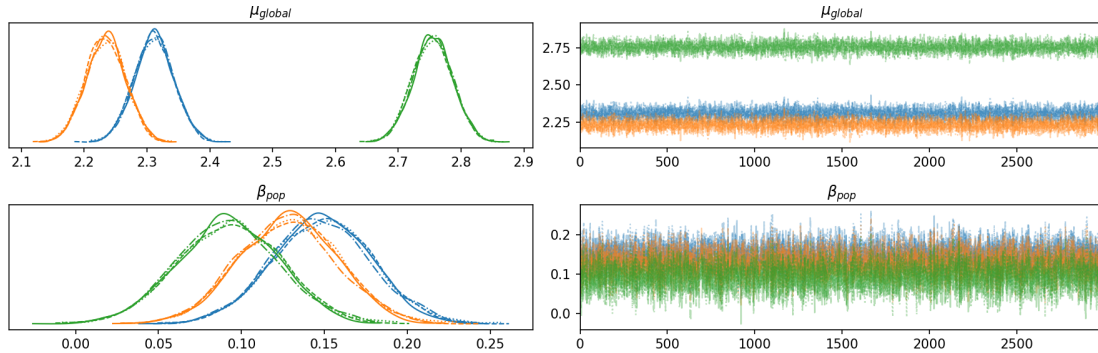


Figure 6: Trace: Dirichlet Pathogen Prevalence Model

The priors all have wider ranged HDIs and all of them contain zero. This indicates that there isn't an encoded directionality for any of them which is a positive sign for the prior parameters.

Table 6: Prior Predictive Summary: Dirichlet Pathogen Prevalence Model

Parameter	Mean	SD	HDI 2.5%	HDI 97.5%
μ_{global} [Influenza A]	0.003	1.020	-2.108	2.036
μ_{global} [RSV]	-0.025	1.019	-2.018	1.892
μ_{global} [SARS-CoV-2]	-0.001	1.027	-1.876	2.046
β_{pop} [Influenza A]	-0.004	0.497	-0.950	0.958
β_{pop} [RSV]	0.001	0.507	-1.062	0.929
β_{pop} [SARS-CoV-2]	-0.005	0.506	-0.922	1.069

Limitations

WastewaterSCAN uses PMMoV normalization but CDC analysis switched to using non-normalized data in August, 2025. The CDC made this switch to decrease the effect of normalization on the changes in viral concentration across time. There is a normalization inconsistency in the structure of the data.

More broadly, the CDC NWSS data comes from a multitude of sources at the local level, state level, and from partnerships with universities and private industry. This creates an inherent flaw within the WVAL calculation, as comparisons across sources are not inherently reliable. Both of these models depend on precise and accurate measure of viral concentration, and can be easily biased or skewed with inconsistency in data collection or aggregation. This might also be able to explain some of the outliers discovered early in the data cleaning process.

References

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